

2.1.2. Dictionary Operations

13:32

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

DictOper...

Submit

```

1  # Initial dictionary with 10 predefined records
2  student = {
3      1: "Amit",
4      2: "Riya",
5      3: "Kiran",
6      4: "Neha",
7      5: "Arjun",
8      6: "Pooja",
9      7: "Rahul",
10     8: "Sneha",
11     9: "Vikram",
12    10: "Anjali"
13 }
14 print("Original Dictionary:", student)
15
16 # Insertion
17 key_insert = int(input())
18 value_insert = input()
19 student[key_insert] = value_insert
20
21 print("After Insertion:", student)
22
23 # Update
24 key_update = int(input())
25 value_update = input()
    
```

Terminal Test cases

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d60/6773e9c2f4ab787eca9d09e4

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.1.1. Calculate Momentum08:52

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 mass = float(input())
2 velocity = float(input())
3 momentum = mass*velocity
4 print(f"{momentum:.2f}kgm/s")
```

TerminalTest cases

Submit

Submit

PrevResetSubmitNext

35°CPartly sunny

Search

ENG IN

21:1807-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.1.2. Conditional Calculation Based on the Number of Digits

10/27

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

```
1 import math
2
3 n=int(input())
4 if 1 <= n <= 9:
5     print(n*n)
6 elif 10 <= n <= 99:
7     print(f"{math.sqrt(n):.2f}")
8 elif 100 <= n <= 999:
9     print(f"{n**(1/3):.2f}")
10 else:
11     print("Invalid")
```

Terminal Test cases

35°C Partly sunny

Search

ENG IN

21:18 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.1.3. Days between Two Dates

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import datetime
2
3 date1 = input().strip()
4 date2 = input().strip()
5
6 d1 = datetime.strptime(date1, "%Y-%m-%d")
7 d2 = datetime.strptime(date2, "%Y-%m-%d")
8 difference = (d2 - d1).days
9 print(difference)
```

Terminal Test cases

29°C Mostly clear

Search

21:19 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.1.4. Reverse a Number

07:58

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2 reverse = 0
3 while n != 0:
4     digit = n % 10
5     reverse = reverse * 10 + digit
6     n = n // 10
7 print(reverse)
```

Terminal Test cases

Prev Reset Submit Next

29°C Mostly clear

Search

ENG IN

21:19 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.1.5. Multiplication Table

01:56

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

```
1 n = int(input())
2
3 for i in range(1,11):
4     print(f"{n} x {i} = {n*i}")
5
6
```

Terminal Test cases

Prev Reset Submit Next

29°C Mostly clear Search 21:19 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.2.1. Pass or Fail

23:03

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

Submit

```
1 # write your code here
2 def calculate_grade():
3     # Input the number of courses
4     num_courses = int(input())
5
6     # Input the marks for each course
7     marks = list(map(int, input().split()))
8
9     # Check if the number of marks entered matches the number of courses
10    if len(marks) != num_courses:
11        print()
12        return
13
14    # Check for failure in any course
15    if any(mark < 40 for mark in marks):
16        print("Fail")
17        return
18
19    # Calculate aggregate percentage
20    total_marks = sum(marks)
21    aggregate_percentage = (total_marks / (num_courses * 100)) * 100 # Assuming each course is out of 100
22
23    # Print the aggregate percentage rounded to two decimal places
24    print(f"Aggregate Percentage: {aggregate_percentage:.2f}")
```

Terminal Test cases

Thunderstorm w... In effect

Search

ENG IN

21:20 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.2.4. Pattern - 2

08:20

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 # Type Content here...
2 def print_triangle_pattern(rows):
3     for i in range(1, rows + 1):
4         # Generate the numbers for the current row
5         for j in range(1, i + 1):
6             print(j, end=' ')
7         # Move to the next line after each row
8         print()
9
10 # Input: number of rows
11 try:
12     num_rows = int(input())
13     if num_rows < 1:
14         print("Please enter a positive integer.")
15     else:
16         print_triangle_pattern(num_rows)
17 except ValueError:
18     print("Invalid input. Please enter an integer.")
19
```

Terminal Test cases

Thunderstorm w... In effect

Search

ENG IN 21:20 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

1.2.3. Pattern - 1

05:16

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Type Content here...
2 def print_triangle(rows):
3     """Function to print a right-angled triangle pattern of
4     asterisks."""
5     for i in range(1, rows + 1):
6         print('*' * i)
7
8 def main():
9     # Get the number of rows from the user
10    rows = int(input())
11
12    # Print the triangle pattern
13    print_triangle(rows)
14
15 if __name__ == "__main__":
16    main()
```

Terminal Test cases

Thunderstorm w...
In effect

Search

ENG
IN

21:20
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

3.2.5. Numpy: Copying and Viewing Arrays

00:37

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Sample Test Cases

copyAnd...

Submit

Debugger

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
16 print("Original array after modifying view:", original_array)
17 print("View array:", view_array)
18
19 # Modify the copy
20 copy_array[1] = 88
21 print("Original array after modifying copy:", original_array)
22 print("Copy array:", copy_array)
23
```

Terminal Test cases

Thunderstorm w... In effect

Search

ENG IN

21:22 07-05-2026

CODETANTRA

HomeLearn Anywhere

202501090111@mitaoe.ac.inSupportLogout

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bit...00:38

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

- Arithmetic Operations:**
 - Compute the element-wise sum, difference, and product of the two arrays.
- Statistical Operations:**
 - Calculate the mean, median, and standard deviation of array A.
- Bitwise Operations:**
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     arr_A = np.array(A)
7     arr_B = np.array(B)
8
9     # Arithmetic Operations
10    sum_result = arr_A + arr_B # Element-wise sum
11    diff_result = arr_A - arr_B # Element-wise difference
12    prod_result = arr_A * arr_B # Element-wise product
13
14    # Statistical Operations
15    mean_A = np.mean(arr_A) # Mean of A
16    median_A = np.median(arr_A) # Median of A
17    std_dev_A = np.std(arr_A) # Standard deviation of A
18
19    # Bitwise Operations
20    and_result = arr_A & arr_B # Bitwise AND
21    or_result = arr_A | arr_B # Bitwise OR
22    xor_result = arr_A ^ arr_B # Bitwise XOR
23
24    # Output results with one space between each element
25    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

TerminalTest cases

Thunderstorm w...In effect

Search

ENGIN

21:2207-05-2026

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Thunderstorm w... In effect Search [Taskbar icons: File Explorer, Edge, etc.] ENG IN 21:22 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting00:40

The given code in the editor takes a single array, `array1`, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- `search_value`: The value to search for in the array.
- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching**: Find the indices where `search_value` appears in `array1` and print these indices.
- Counting**: Count how many times `count_value` appears in `array1` and print the count.
- Broadcasting**: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- Sorting**: Sort `array1` in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing `array1`.
- An integer `search_value` represents the value to search for in the array.
- An integer `count_value` represents the value to count in the array.
- An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

- The indices where `search_value` occurs in `array1`.
- The count of occurrences of `count_value` in `array1`.
- The array after adding the `broadcast_value` to each element.
- The sorted array.

arrayOpe...

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split()))))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
12
13 # Count occurrences in array1
14
15 # Broadcasting addition
16
17 # Sort the first array
18 # Find indices where search_value matches in array1
19 search_indices = np.where(array1 == search_value)[0]
20 print( search_indices)
21
22 # Count occurrences of count_value in array1
23 count_occurrences = np.count_nonzero(array1 == count_value)
24 print( count_occurrences)
25
```

Terminal Test cases

< Prev Reset Submit Next >

Hot days ahead 29°C

Search

ENG IN 21:23 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.4. Most Frequently Sold Product Pairs

00:44

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B
- 2025-01-05: Product A, Product C

frequentl...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
12 grouped = df.groupby("Date")["Product"].apply(list)
13 pairs_counter = Counter()
14 for products in grouped:
15     pairs = combinations(sorted(products), 2)
16     pairs_counter.update(pairs)
17 max_frequency = max(pairs_counter.values())
18 most_common_pairs = [(pair, freq) for pair, freq in
19     pairs_counter.items() if freq == max_frequency]
20 # Output the most frequent product pairs
21 for (product1, product2), frequency in most_common_pairs:
22     print(f"{product1} and {product2}: {frequency} times")
23
24 # Output the most frequent product pairs
```

Terminal Test cases

Hot days ahead 29°C

Search

ENG IN 21:24 07-05-2026

5.2.5. Bar Plot for Survival by Pclass

00:38

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Pclass"** to the chart.
- Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 survival_counts = data.groupby('Pclass')
18 ['Survived'].value_counts().unstack().fillna(0)
19 survival_counts.plot(kind='bar', stacked=True)
20 plt.title('Survival by Pclass')
21 plt.xlabel('Pclass')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

Terminal Test cases

Prev

Reset

Submit

Next

5.2.11. Scatter Plot for Age vs. Fare by Survived00:43

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

- Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
- Set the title of the plot to **"Age vs. Fare by Survival"**.
- Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, , S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.2833, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, , S
4, 1, 1, "Futrelle, Mrs. Jacques Heath (Lily May Peel)", female, 35, 1, 0, 113803, 53.1, C123, S
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 373450, 8.05, , S
```

AgeFareS...Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Scatter Plot for Age vs. Fare by Survived
17 colors = data['Survived'].map({0: 'red', 1: 'blue'})
18 plt.scatter(data['Age'], data['Fare'], c=colors)
19 plt.title('Age vs. Fare by Survival')
20 plt.xlabel('Age')
21 plt.ylabel('Fare')
22 plt.show()
23
24
```

Terminal Test cases

5.2.10. Scatter Plot for Age vs. Fare

00:47

⌵ ⌵ ⌵ ⌵ ⌵

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

AgeFareS...

Submit

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Box Plot for Fare by Pclass

17 plt.figure()

18 plt.scatter(data['Age'], data['Fare'])

19 plt.title('Age vs. Fare')

20 plt.xlabel('Age')

21 plt.ylabel('Fare')

22 plt.show()

23

24

25

Terminal Test cases

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e2a1727c0b912d860eb39b

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.9. Box Plot for Fare by Pclass

00:34

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McGowan, Mr. Thomas J",male,54,0,0,57463,51.8025,546,C

BoxPlotF...

Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Fare',by='Pclass')
19 plt.title('Fare by Pclass')
20 plt.suptitle('')
21 plt.xlabel('Pclass')
22 plt.ylabel('Fare')
23 plt.show()
24
25

Terminal Test cases

29°C Mostly clear

Search

21:28 07-05-2026

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e29db57c0b912d860eab45

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.8. Box Plot for Age by Survived

00:48

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McGowan, Mr. Thomas J",male,54,0,0,57463,51.8025,E46,C

BoxPlotF...

Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Age', by='Survived')
19 plt.title('Age by Survival')
20 plt.suptitle('')
21 plt.xlabel('Survived')
22 plt.ylabel('Age')
23 plt.show()
24
25

Terminal Test cases

29°C Mostly clear

Search

21:28 07-05-2026

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e2992e7c0b912d860ea071

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.7. Box plot for Age Distribution

00:32

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- Use the **Pclass** column to group the data for the boxplot.
- Set the title of the plot to **"Age by Pclass"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

BoxPlotF...

Submit

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Pclass
17 data.boxplot(column = 'Age', by='Pclass')
18 plt.title('Age by Pclass')
19 plt.suptitle('')
20 plt.xlabel('Pclass')
21 plt.ylabel('Age')
22 plt.show()
23
24

Terminal Test cases

29°C Mostly clear

Search

21:28 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.6. Bar Plot for Survival by Embarked

01:51

AA

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked**" to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17 survival_counts = data.groupby('Embarked_Q')
18 ['Survived'].value_counts().unstack().fillna(0)
19 survival_counts.plot(kind = 'bar', stacked = True)
20 plt.title('Survival by Embarked')
21 plt.xlabel('Embarked')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

Terminal Test cases

< Prev

Reset

Submit

Next >

29°C Mostly clear

Search

ENG IN

21:27 07-05-2026

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e28fb27c0b912d860e7db4

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.4. Bar Plot for Survival by Gender

00:32

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Gender"** to the chart.
- Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
```

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Gender
17
18 survival_counts = data.groupby('Sex')
19 ['Survived'].value_counts().unstack()
20 survival_counts.plot(kind='bar', stacked=True)
21 plt.title('Survival by Gender')
22 plt.xlabel('Gender')
23 plt.ylabel('Count')
24 plt.legend(['Not Survived', 'Survived'])
25 plt.show()
```

Terminal

Test cases

Prev

Reset

Submit

Next

29°C Mostly clear

Search

21:27 07-05-2026

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e28d6e7c0b912d860e75ad

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.3. Bar plot of survival rate of passengers

00:33

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title "Survival Count" to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

BarPlotOf...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Bar Plot for Survival Rate

17 survival_counts = data['Survived'].value_counts()

18 survival_counts.plot(kind='bar')

19 plt.title('Survival Count')

20 plt.xlabel('Survived')

21 plt.ylabel('Count')

22 plt.show()

23

Terminal

Test cases

Prev

Reset

Submit

Next

Tomorrow's high
To tie record

Search

ENG IN

21:26
07-05-2026

16 WhatsApp

Course

https://mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d6b/698c54860aba96214c8b7d6d/67e28a397c0b912d860e689c

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.2. Histogram of passenger information of Titanic00:41

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use 30 bins for the histogram.

2. Set the edge color of the bars to black (k).

3. Label the x-axis as 'Age' and the y-axis as 'Frequency'.

4. Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McGowan, Mr. Thomas J",male,54,0,0,57463,51.8025,E46,C

Histogram...

1import pandas as pd

2import matplotlib.pyplot as plt

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# Data Cleaning

8data['Age'].fillna(data['Age'].median(), inplace=True)

9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10data.drop('Cabin', axis=1, inplace=True)

11

12# Convert categorical features to numeric

13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16# Write your code here for Histogram

17# Write your code here for Histogram

18plt.hist(data['Age'],bins=30,edgecolor='k')

19plt.title('Age Distribution')

20plt.xlabel('Age')

21plt.ylabel('Frequency')

22plt.show()

23

Terminal

Test cases

Prev

Reset

Submit

Next

Tomorrow's high
To tie record

Search

ENG
IN

21:26
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.2.1. Titanic Dataset

02:21

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.

titanicDat...

Submit

1 `import pandas as pd`

2 `import matplotlib.pyplot as plt`

3

4 `# Load the Titanic dataset from the CSV file`

5 `df = pd.read_csv('titanic.csv')`

6

7 `# Set up the figure for 5 subplots`

8 `fig, axes = plt.subplots(3, 2, figsize=(12, 12))`

9

10 `# write the code..`

11 `count=df['Pclass'].value_counts().sort_index()`

12 `count_gen=df['Gender'].value_counts()`

13 `count_s=df['Survived'].value_counts().sort_index()`

14

15 `axes[0,0].bar(count.index,count.values,color=['skyblue'])`

16 `axes[0,0].set_title('Passenger Class Distribution')`

17 `axes[0,0].set_xlabel('Pclass')`

18 `axes[0,0].set_ylabel('Count')`

19

20 `axes[0,1].pie(count_gen.values,labels=['male','female'],colors=`

21 `['lightblue','lightcoral'],autopct='%1.1f%%')`

22 `axes[0,1].set_title('Gender Distribution')`

23

24 `axes[1,0].hist(df['Age'],color=['lightgreen'],edgecolor='black',bins=8)`

25 `axes[1,0].set_title('Age Distribution')`

Terminal Test cases

Prev Reset Submit Next

Tomorrow's high
To tie record

Search

ENG IN

21:26
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

5.1.1. Stacked Plot

00:50

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June',
7             'July', 'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 # Write your code...
15 df = pd.DataFrame(data)
16 plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'], df['City_C_Temperature'])
17 plt.xlabel('Month')
18 plt.ylabel('Temperature')
19 plt.title('Temperature Variation')
20 plt.legend(loc='upper left')
21 plt.show()
```

Terminal Test cases

Prev Reset Submit Next

Tomorrow's high
To tie record

Search

ENG IN

21:26
07-05-2026

CODETANTRA

Home

Learn Anywhere

202501090111@mitaoe.ac.inSupportLogout

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

00:28

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

Pas sen ger id	Sur vive d	Pcl ass	Na me	Sex	Age	Sib Sp	Par ch	Tick et	Far e	Cab in	Em bar ked

Sample Data:

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9 # 1. Get the number of survivors by gender
10 print(data[data['Survived']==1]['Sex'].value_counts())
11
12 # 2. Get the number of non-survivors by gender
13 print(data[data['Survived']==0]['Sex'].value_counts())
14
15 # 3. Get the number of survivors by embarked location
16 print(data[data['Survived']==1]['Embarked_S'].value_counts())
17 # 4. Get the number of non-survivors by embarked location
18 print(data[data['Survived']==0]['Embarked_S'].value_counts())
19
20 # 5. Calculate the percentage of children (Age < 18) who survived
21 print(data[data['Age']<18]['Survived'].mean())
22
23 # 6. Calculate the percentage of adults (Age >= 18) who survived
24 print(data[data['Age']>17]['Survived'].mean())
25
```

TerminalTest cases

PrevResetSubmitNext

Air: Moderate Now

Search

ENG IN

21:25 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.7. Titanic Dataset Analysis and Data Cleaning - 301:23

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9 # 1. Calculate the survival rate by class
10 print(data.groupby('Pclass')['Survived'].mean())
11
12 # 2. Calculate the survival rate by embarked location
13 print(data.groupby('Embarked_S')['Survived'].mean())
14
15 # 3. Calculate the survival rate by family size
16 print(data.groupby('FamilySize')['Survived'].mean())
17
18 # 4. Calculate the survival rate by being alone
19 print(data.groupby('IsAlone')['Survived'].mean())
20
21 # 5. Get the average fare by class
22 print(data.groupby('Pclass')['Fare'].mean())
23
24 # 6. Get the average age by class
25 print(data.groupby('Pclass')['Age'].mean())
```

Terminal Test cases

Prev Reset Submit Next

Air: Moderate Now

Search

ENG IN 21:25 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.6. Titanic Dataset Analysis and Data Cleaning - 20101

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 # 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
10
11 # 2. Convert 'Sex' to numeric (male: 0, female: 1)
12 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
13
14 # 3. One-hot encode the 'Embarked' column
15 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
16
17 # 4. Get the mean age of passengers
18 print(data['Age'].mean())
19
20
21
22 # 5. Get the median fare of passengers
23 print(data['Fare'].median())
24
25
```

Terminal Test cases

< Prev

Reset

Submit

Next >

Air: Moderate Now

Search

ENG IN

21:25 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.3. City that Sold the Most Products

00:46

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 city_sales = df.groupby('City')['Quantity'].sum().reset_index()
11 best_city_row = city_sales.loc[city_sales['Quantity'].idxmax()]
12 best_city = best_city_row['City']
13 # Display the result
14 print(f"City sold the most products: {best_city}")
15
```

Terminal

Test cases

Hot days ahead
29°C

Search

ENG IN

21:24
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.5. Titanic Dataset Analysis and Data Cleaning

01:11

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

- Display the first 5 rows of the dataset.
- Display the last 5 rows of the dataset.
- Get the shape of the dataset (number of rows and columns).
- Get a summary of the dataset (using `.info()`).
- Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
- Check for missing values and display the count of missing values for each column.
- Fill missing values in the 'Age' column with the median age.
- Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
- Drop the 'Cabin' column due to many missing values.
- Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

titanicDat...

Submit

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # 1. Display the first 5 rows of the dataset
8 print(data.head())
9 # 2. Display the last 5 rows of the dataset
10 print(data.tail())
11
12 # 3. Get the shape of the dataset
13 print(data.shape)
14
15 # 4. Get a summary of the dataset (info)
16 print(data.info())
17
18 # 5. Get basic statistics of the dataset
19 print(data.describe())
20
21 # 6. Check for missing values
22 print(data.isnull().sum())
23
24 # 7. Fill missing values in the 'Age' column with the median age
25 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Terminal Test cases

Prev Reset Submit Next

Air: Moderate Now

Search

ENG IN 21:25 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.1.2. Dictionary to dataframe

03:38

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

datafram...

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17 # Adding a new row
18 name=input("New name: ")
19 age=int(input("New age: "))
20 new={'Name': name, 'Age': age}
21 df=df.append(new,ignore_index=True)
22
23 # Display the DataFrame after adding a new row
24 print("After adding a row:\n",df)
25
```

Terminal Test cases

< Prev

Reset

Submit

Next >

Hot days ahead
29°C

Search

ENG IN

21:23
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.1. Month with the Highest Total Sales

01:07

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

Submit

1 import pandas as pd

2

3 # Prompt the user for the file name

4 file_name = input()

5

6 # Load the data

7 df = pd.read_csv(file_name)

8 df["Date"] = pd.to_datetime(df["Date"])

9 df["Month"] = df["Date"].dt.to_period("M")

10 df["Total Sales"] = df["Quantity"] * df["Price"]

11 monthly_sales = df.groupby("Month")["Total Sales"].sum()

12

13 # Find the month with the highest total sales

14 best_month = monthly_sales.idxmax()

15 highest_sales = monthly_sales.max()

16

17 print(f"Best month: {best_month}")

18 print(f"Total sales: \${highest_sales:.2f}")

19

Terminal Test cases

Hot days ahead 29°C

Search

21:24 07-05-2026

CODETANTRA

HomeLearn Anywhere

202501090111@mitaoe.ac.inSupportLogout

4.1.3. Student Information

00:48

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

Submit

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age",
6 "Grade"])
7
8
9 # write your code here..
10
11 print("First five rows:")
12 print(data.head())
13
14 avg_age=round(data["Age"].mean(),2)
15 print(f"Average age: {avg_age}")
16
17 print("Students with a grade up to B")
18 fs=data[data["Grade"]<="B"]
19 print(fs)
20
```

TerminalTest cases

Hot days ahead
29°C

Search

21:24
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.2.2. Best Selling Product

01:54

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

Submit

1 import pandas as pd

2

3 # Prompt the user for the file name

4 file_name = input()

5

6 # Load the data

7 df = pd.read_csv(file_name)

8

9 product_sales = df.groupby('Product')['Quantity'].sum()

10 # Find the product with the highest total quantity sold

11 best_product = product_sales.idxmax()

12 highest_quantity = product_sales.max()

13

14 # Display the result

15 print(f"Best selling product: {best_product}")

16 print(f"Total quantity sold: {highest_quantity}")

17

Terminal Test cases

Hot days ahead 29°C

Search

ENG IN

21:24 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

4.1.1. Pandas - series creation and manipulation

01:17

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 series = pd.Series(numbers)
7
8 # Create a new series to indicate whether each number is even (True) or
  odd (False)
9 is_even = series % 2 == 0
10 grouped = series.groupby(is_even).mean()
11
12 # Display the mean of even and odd numbers with labels
13 grouped.index = ['Even' if is_even else 'Odd' for is_even in
  grouped.index]
14 print("Mean of even and odd numbers:")
15 print(grouped)
16
```

Terminal Test cases

Hot days ahead
29°C

Search

ENG
IN

21:23
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

3.1.1. Numpy Array Operations

00:53

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

1 import numpy as np

2

3 rows, cols = map(int, input().split())

4 if rows > 0 and cols > 0:

5 elements = []

6 for i in range(rows):

7 elements.extend(map(int, input().split()))

8 arr = np.array(elements).reshape(rows, cols)

9 else: arr = np.array([]).reshape(rows, cols)

10 print(arr)

11 print(arr.ndim)

12 print(arr.shape)

13 print(arr.size)

14

Terminal Test cases

Thunderstorm w...
In effect

Search

ENG
IN

21:21
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

3.2.1. Numpy: Matrix Operations

07:12

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)
2. Subtraction (`matrix_a - matrix_b`)
3. Element-wise Multiplication (`matrix_a * matrix_b`)
4. Matrix Multiplication (`matrix_a · matrix_b`)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

matrixOp...

Submit

1 import numpy as np

2

3 # Input matrices

4 print("Enter Matrix A:")

5 matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

6

7 print("Enter Matrix B:")

8 matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

9

10

11 # Addition

12 print("Addition (A + B):")

13 print(matrix_a + matrix_b)

14 # Subtraction

15 print("Subtraction (A - B):")

16 print(matrix_a - matrix_b)

17 # Multiplication (element-wise)

18 print("Element-wise Multiplication (A * B):")

19 print(matrix_a * matrix_b)

20 # Matrix multiplication (dot product)

21 print("A dot B:")

22 print(matrix_a @ matrix_b)

23 # Transpose

24 print("Transpose of A:")

25 print(matrix_a.T)

Terminal Test cases

Thunderstorm w...
In effect

Search

ENG
IN

21:22
07-05-2026

CODETANTRA

Home

Learn Anywhere

202501090111@mitaoe.ac.inSupportLogout

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays00:33

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (side by side)
11 horizontal_stack = np.hstack((arr1, arr2))
12 print("Horizontal Stack:")
13 print(horizontal_stack)
14
15 # Perform vertical stacking (one below the other)
16 vertical_stack = np.vstack((arr1, arr2))
17 print("Vertical Stack:")
18 print(vertical_stack)
19
```

TerminalTest cases

Thunderstorm w...
In effect

Search

ENG
IN

21:22
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

3.2.7. Student Data Analysis and Operations

02:22

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who not less than 40 marks in Subject 1:** Count the number of

Operatio...

```
1 import numpy as np
2
3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4 # 1. Print all student details
5 print("All student Details:\n",a)
6
7 # 2. print total students
8
9 print("Total Students:", len(a))
10
11 # 3. Print all student Roll numbers
12 print("All Student Roll Nos", a[:,0] ...)
13
14 # 4. Print subject 1 marks
15 print("Subject 1 Marks", a[:,1])
16
17 # 5. print minimum marks of Subject 2
18 print("Min marks in Subject 2", a[:,2].min() ...)
19
20 # 6. print maximum marks of Subject 3
21 print("Max marks in Subject 3", a[:,3].max() ...)
22
23 # 7. Print All subject marks
24 a1 = np.delete(a,0,1)
25 print("All subject marks:", a1 ...)
```

Terminal Test cases

< Prev Reset Submit Next >

Hot days ahead 29°C

Search

ENG IN

21:23 07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

2.2.2. Captain of the Team

00:14

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases

captainof...

Submit

1

Read 11 integers from input

2

heights = list(map(int, input().split()))

3

4

Find the tallest player

5

tallest = max(heights)

6

7

Print the height of the tallest player

8

print(tallest)

9

Terminal

Test cases

Thunderstorm w...
In effect

Search

ENG
IN

21:21
07-05-2026

CODETANTRA

Home

Learn Anywhere

202501090111@mitaoe.ac.inSupportLogout

1.2.2. Fibonacci Series28:47

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2
3     if n==1:
4         return 0
5     elif n == 2:
6         return 1
7     else:
8         return fibonacci(n - 1) + fibonacci(n - 2)
9
10
11 n = int(input())
12 for i in range(1, n + 1):
13     print(fibonacci(i), end=" ")
14
15
```

TerminalTest cases

Thunderstorm w...
In effect

Search

ENG
IN

21:20
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

2.1.1. List operations04:59

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Sample Test Cases

listOps.py

```
1 # write the code..
2 def display_menu():
3     print("1. Add")
4     print("2. Remove")
5     print("3. Display")
6     print("4. Quit")
7
8 def main():
9     integer_list = []
10
11     while True:
12         display_menu()
13         choice = input("Enter choice: ")
14
15         if choice == '1': # Add
16             try:
17                 num = int(input("Integer: "))
18                 integer_list.append(num)
19                 print(f"List after adding: {integer_list}")
20             except ValueError:
21                 print("Invalid input")
22
23         elif choice == '2': # Remove
24             if not integer_list:
25                 print("List is empty")
```

Terminal Test cases

Prev Reset Submit Next

Thunderstorm w...
In effect

Search

ENG
IN

21:21
07-05-2026

CODETANTRA

Home Learn Anywhere

202501090111@mitaoe.ac.in Support Logout

2.1.2. Dictionary Operations

11:46

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

DictOper...

```
1 # Initial dictionary with 10 predefined records
2 student = {
3     1: "Amit",
4     2: "Riya",
5     3: "Kiran",
6     4: "Neha",
7     5: "Arjun",
8     6: "Pooja",
9     7: "Rahul",
10    8: "Sneha",
11    9: "Vikram",
12   10: "Anjali"
13 }
14
15 # Display original dictionary
16 print("Original Dictionary:", student)
17
18 # Insertion
19 insert_key = int(input())
20 insert_value = input()
21 student[insert_key] = insert_value
22 print("After Insertion:", student)
23
24 # Update
25 update_key = int(input())
```

Terminal Test cases

Thunderstorm w...
In effect

Search

ENG IN

21:21
07-05-2026

CTP1709...

Submit

1

Read the array of integers

2

arr = list(map(int, input().split()))

3

4

Read the key element to search

5

key = int(input())

6

7

Perform linear search

8

found = False

9

for i in range(len(arr)):

10

if arr[i] == key:

11

print(i)

12

found = True

13

break

14

15

If element not found

16

if not found:

17

print("Not found")

18

Terminal

Test cases

< Prev

Reset

Submit

Next >

2.2.1. Linear search Technique

00:15

AA

🌙

📄

🔗

—

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:

Input:

1 2 3 4 3 5 6

3

Output:

2

Sample Test Cases

+

Thunderstorm w...

In effect

Search

ENG

IN

21:21

07-05-2026